KERNELS AND COKERNELS OF MAPS OF QUASI-COHERENT SHEAVES ON AFFINES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1642-1649, following the statement at 1635-1640.

Editorial note. The kernel/cokernel identifications are displayed parenthetically; exactness of tilde on stalks (localization) is the standard ingredient invoked by 'the exactness of the functor tilde'.

Allowed standard prerequisites (editorial).

- The two tilde constructions agree. lemma-compare-constructions (card L26415, cited at ~1644). Every map of tildes comes from a module map.

Source reference. Original lemma 01I7: sources/stacks-schemes/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X = \mathrm{Spec}(R)$ be an affine scheme. Kernels and cokernels of maps of quasi-coherent $\mathcal{O}_X$-modules are quasi-coherent.*

Proof. This follows from the exactness of the functor $\widetilde{\ }$ since by Lemma 01I7 we know that any map $\psi : \widetilde{M} \to \widetilde{N}$ comes from an R-module map $\varphi : M \to N$. (So we have $\mathrm{Ker}(\psi) = \widetilde{\mathrm{Ker}(\varphi)}$ and $\mathrm{Coker}(\psi) = \widetilde{\mathrm{Coker}(\varphi)}$.) □

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